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COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

ByteQuest: A Gamified Programming Learning Platform Using Comparative Machine Learning Models for Predicting Programming-Task Performance

An Undergraduate Thesis
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In Partial Fulfillment
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CHAPTER III

RESEARCH METHODOLOGY

This chapter presents the proposed methodology for developing and evaluating ByteQuest. It discusses the research design, research locale, research respondents, data-gathering procedure, algorithm design, research instruments, statistical treatment, materials, schedule and timeline, project-team responsibilities, and budget management plan. Activities that have not yet been completed are presented as planned procedures.

Research Design

The study will employ a research and development approach to guide the design, development, and evaluation of ByteQuest, a gamified programming-learning platform for selected Computer Science students at North Eastern Mindanao State University (NEMSU). Development will follow an Agile iterative-incremental software development life cycle (SDLC), in which system features will be implemented, tested, and refined in successive iterations.

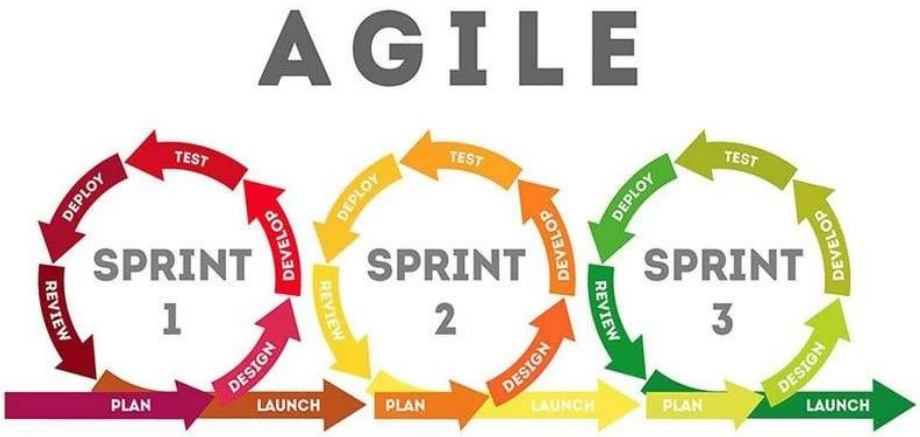


Figure 1. *Agile iterative-incremental software development life cycle*

The planned platform will provide quizzes and programming challenges in Python, C++, and Java, together with game elements such as points, badges, quests, and progress indicators. It will also include an AI-assisted content-generation feature that retrieves relevant passages from uploaded programming-related PDF materials and supplies them to a large language model to help generate quizzes, review questions, and programming challenges. Generated materials will be reviewed for correctness, relevance, and alignment with the source material before being made available for use.

The programming-assessment component will evaluate submitted programs against predefined test cases. The resulting task outcomes and other performance information available before a target task will be prepared as predictor features for two supervised classification models: Logistic Regression

and Random Forest. The models will be compared in predicting whether a student will successfully complete a subsequent programming task. The study will evaluate the platform in terms of functionality and usability and will evaluate the models using accuracy, precision, recall, and F1-score. The study will not assume in advance that one model will outperform the other.

The study is not designed to establish that ByteQuest improves students' learning outcomes or motivation. Its planned evaluation focuses on system functionality and usability and on the predictive performance of the two classification models.

Research Locale

The study will be conducted at North Eastern Mindanao State University (NEMSU) Main Campus in Tandag City, Surigao del Sur. The locale is selected because ByteQuest is being designed for the programming-learning context of selected students enrolled in the university's Bachelor of Science in Computer Science program. The specific classes, laboratories, or other locations involved in recruitment and evaluation will be identified after coordination with the appropriate university offices and instructors.



***Figure 2**. Map of NEMSU Tandag*

Research Respondents

The intended respondents will include selected Bachelor of Science in Computer Science students and programming instructors at NEMSU. Students will be invited to participate in the pre-development needs assessment and, subject to the approved study procedure, may also participate in the planned system pilot and evaluation. Programming instructors will be invited to provide

information about programming-teaching challenges and to participate in relevant system evaluation activities.

The proposed inclusion criteria for student respondents are: (1) currently enrolled in the Bachelor of Science in Computer Science program at NEMSU; and (2) have relevant experience with one or more of the programming languages covered by ByteQuest—Python, C++, or Java. Instructor eligibility will be based on teaching or relevant instructional experience in programming, subject to confirmation with the adviser.

The sampling method and target sample size have not yet been finalized. They will be established before formal recruitment, in consultation with the thesis adviser and in accordance with the eligible population, study requirements, participant availability, and any applicable institutional procedures. The number of participants and the amount of usable programming-task data will be reported accurately after the approved data-collection process. Participation will be voluntary and subject to the required permission and consent procedures.

Data Gathering Procedure

Data gathering will proceed in stages. At the time this draft was prepared, the researchers had conducted an informal preliminary interview with seven

Computer Science students from different year levels. Formal questionnaire validation, permission letters, and formal pre-development data gathering had not yet been completed.

Stage 1: Preliminary Consultation (Completed)

The researchers conducted informal interviews with seven Computer Science students from different year levels to obtain exploratory insights into programming-related learning experiences, debugging practices, and challenges. The consultation helped the researchers identify issues to consider during system planning. It was exploratory and will not be treated as the formal needs assessment or as evidence that all Computer Science students experience the same challenges.

Stage 2: Research Instrument Validation (Pending)

The student and instructor questionnaires will be submitted to appropriate validators for review. The review will consider whether the items are clear, relevant, and aligned with the study's objectives. The researchers will revise the instruments based on the validators' comments. Any additional pilot testing or reliability analysis will be conducted only if required by the adviser, validators, or applicable institutional guidance. The questionnaires will not be

administered as the formal needs-assessment instruments until the required validation and revisions have been completed.

Stage 3: Permission and Coordination (Pending)

The researchers will prepare and submit the required request letters to the appropriate university offices and coordinate with relevant instructors before recruiting respondents or collecting formal research data. Participants will receive information about the purpose of the study, the activities involved, the voluntary nature of participation, and how their responses or performance data will be handled. Required consent and institutional procedures will be followed before data collection.

Stage 4: Formal Pre-development Needs Assessment (Pending)

After the questionnaires have undergone the required validation and the necessary permissions have been secured, the researchers will administer the student and instructor questionnaires to eligible respondents. The student questionnaire will gather information about learning experiences, syntax and programming challenges, practice habits, debugging approaches, progress tracking, and the perceived usefulness of proposed ByteQuest features. The instructor questionnaire will gather information about programming-teaching

challenges, student practice and persistence, debugging and progress monitoring, and the potential usefulness of the proposed platform. Responses will be organized for analysis and will inform requirements refinement.

Stage 5: System Development and Pilot Performance-Data Collection (Planned)

The researchers will develop ByteQuest through iterative implementation and testing. Once the relevant programming-task and assessment features are ready and the required approvals are in place, participating students will complete selected programming activities during a planned pilot. The system is intended to record task submissions and their outcomes as determined by predefined test cases. These records will be reviewed and prepared for the machine-learning component.

For each prediction example, predictor features will be drawn only from activities completed before the target programming task. The target label will represent whether the student successfully completed that subsequent task according to the study's assessment rules. Compilation errors and runtime errors will be handled according to the defined task-assessment rules. Infrastructure or execution-service failures will be distinguished from student

performance and will not automatically be labeled as unsuccessful task outcomes.

The feasibility of model training will depend on the quantity, quality, and distribution of usable task records produced during the approved pilot. The researchers will document the final dataset size and class distribution and will avoid claiming that the models are adequately evaluated if the collected data do not support a meaningful evaluation. The final data-splitting method and its rationale will be specified before model evaluation to reduce information leakage, particularly because one student may contribute multiple task records.

Stage 6: System and Model Evaluation (Planned)

After implementation, ByteQuest will undergo testing to determine whether its features operate according to the specified requirements. Intended users will also be invited to evaluate the system's functionality and usability using the approved evaluation procedure and instrument. Separately, the Logistic Regression and Random Forest models will be evaluated using the prepared programming-performance dataset and the selected evaluation procedure. The results will be reported without presuming that either model or the system will achieve a particular outcome.

Algorithm Design

The algorithm design will describe the main components of ByteQuest and how data move through the system. The figures for this section are still to be prepared and should be aligned with the features and technologies actually selected for implementation.

System Architecture

The proposed architecture will show the student and instructor interfaces, application backend, database, programming-task submission and assessment component, AI-assisted content-generation component, and machine-learning prediction component. The RAG workflow will show how text from uploaded programming-related PDF materials is processed, relevant passages are retrieved, and those passages are supplied to the selected language model. Generated content will remain subject to review before use. The architecture should distinguish internal ByteQuest components from external services.

System Flowcharts

The system flowcharts will illustrate the main student and instructor processes. The student flow should include accessing learning activities, answering quizzes or attempting programming tasks, submitting code, receiving the result of test-case assessment, and viewing progress information. The instructor flow should show the relevant content-management, activity-review, and progress-monitoring functions that are included in the final system.

Machine-Learning Workflow

The machine-learning workflow will show performance-data collection, data cleaning and preparation, construction of predictor features from prior activities, definition of the subsequent-task outcome, data splitting, training of Logistic Regression and Random Forest, and evaluation using the selected classification metrics. Features that reveal the target task's outcome or information generated after that outcome will not be used as predictors. The final workflow must identify the actual data-splitting strategy selected for the collected dataset.

Research Instruments

The study will use structured questionnaires for the formal pre-development needs assessment. The questionnaires are still pending validation and will be revised in response to the validators' feedback before formal administration.

Student Questionnaire

The student questionnaire is intended to gather information about students' programming-learning experiences, syntax and programming challenges, practice habits and persistence, debugging approaches, progress tracking, and the perceived potential usefulness of ByteQuest features.

Instructor Questionnaire

The instructor questionnaire is intended to gather information about challenges in teaching programming, observations about student practice and persistence, debugging and progress monitoring, and the potential usefulness of ByteQuest for preparing practice activities and monitoring student progress.

Response Scale and Validation

Both questionnaires use a five-point Likert-type response scale ranging from Strongly Disagree (1) to Strongly Agree (5), as currently drafted. The

instruments will be reviewed by appropriate validators for clarity, relevance, and alignment with the study objectives. The researchers will document the validation feedback and revisions. The questionnaires' validity or reliability will not be claimed before the required review and any required analyses have been completed.

A separate, appropriate procedure will be identified for evaluating the completed system's functionality and usability. The needs-assessment questionnaires will not automatically be treated as a validated system-usability instrument. The final evaluation method and instrument will be confirmed with the adviser before the system evaluation.

Statistical Treatment

The analysis will depend on the type of data collected and the objectives addressed. The methods below are proposed and will be finalized with the adviser before formal analysis.

Needs-Assessment Questionnaire

Frequency counts and percentages will be used to summarize categorical respondent-profile information and response distributions. Weighted means may be used to summarize Likert-type items when appropriate to the instrument and the approved analysis plan. The scoring and

interpretation ranges for weighted means, if used, will be stated in the final methodology and applied consistently.

System Evaluation

The results of the approved functionality and usability evaluation will be summarized using the analysis appropriate to the final evaluation instrument and its response format. If rating-scale responses are collected, response frequencies, percentages, and suitable summary statistics may be reported. The final procedure will identify the specific instrument and scoring approach; no usability or functionality result will be claimed before testing is conducted.

Machine-Learning Evaluation

The Logistic Regression and Random Forest models will be evaluated using a confusion matrix and the classification metrics accuracy, precision, recall, and F1-score. The same target definition and comparable evaluation data will be used for both models. The positive class will be explicitly defined according to the target outcome, and the selected data-splitting strategy will be documented. The results will be used to compare the models; the study will not assume in advance that one will perform better.

Materials

The hardware and software used for the study will be confirmed as development proceeds. The final manuscript should list only resources that are actually available or have been selected for use.

Hardware

The researchers will use available computers and standard peripherals for system development, testing, documentation, and data analysis. The final hardware description will identify the actual development equipment and relevant specifications if these are needed to explain the work. A dedicated GPU will not be claimed as a requirement unless the final implementation demonstrates that it is necessary.

Software and Services

The proposed system requires a frontend, backend/API services, a database, a programming-code execution and test-case assessment environment, a machine-learning implementation environment, and an AI service for content generation. The final choices should be confirmed before submission.

Component	Current plan / decision to confirm

Frontend	React is listed in the current draft; confirm before finalizing.
Backend and APIs	Node.js/Express and Python FastAPI are listed in the current draft; confirm which services each will provide.
Database	TiDB Cloud is listed in the current draft; confirm the selected database and deployment plan.
AI-assisted content generation	Gemini 2.5 Flash is proposed in Chapter I, subject to confirming availability through the selected API service at deployment.
RAG / document processing	Select and document the libraries or services that will extract PDF text, retrieve relevant passages, and pass context to the language model.
Code execution	Specify the selected controlled execution environment and the measures used to isolate submitted code.

Machine learning	Specify the language and libraries actually used to train and evaluate Logistic Regression and Random Forest.
Development and version control	Visual Studio Code and Git are listed in the current draft; confirm actual use.

The code-execution environment is an important system component because student submissions must be executed in a controlled manner. The final methodology should explain, at an appropriate level, how execution is isolated, how time or resource limits are handled, and how infrastructure failures are distinguished from student task outcomes. The selected services and their costs will also be reflected in the budget.

Schedule and Timeline

The project schedule will be presented in a Gantt chart after the group confirms its academic deadlines and the expected duration of each activity. The sequence should reflect dependencies between activities: questionnaire validation and required permissions must precede formal needs-assessment data collection, and pilot performance-data collection must follow the relevant implementation and approval steps.

Activity	**Planned period**	**Status / dependency**
Preliminary consultation	Completed before formal data gathering	Exploratory only
Questionnaire validation and revision	To be scheduled	Pending
Permission letters and coordination	To be scheduled	Pending; before formal recruitment/data collection
Formal pre-development needs assessment	To be scheduled	After required validation and permissions
Requirements analysis and system design	To be scheduled	Refine after needs assessment and requirements review
Iterative system development	To be scheduled	Across planned development iterations

Integration of RAG, code assessment, and database components	To be scheduled	After relevant components are ready
Pilot performance-data collection	To be scheduled	After implementation and required approvals
Dataset preparation and model evaluation	To be scheduled	Depends on sufficient usable records
Functionality and usability evaluation	To be scheduled	After the relevant system features are ready
Documentation and final revisions	To be scheduled	Throughout the project and at completion

Project Teams and Responsibilities

The project will be undertaken collaboratively by Dagohoy, Loren, Mangin, and Somogba. The specific assignments below should be confirmed by the group and revised to reflect the actual division of work. Members may share responsibilities where appropriate.

Team member	**Primary responsibility**	**Specific tasks**
Dagohoy, Venz Andrew P.	To be confirmed by the group	Specify agreed tasks and deliverables.
Loren, Leandro Kieser Y.	To be confirmed by the group	Specify agreed tasks and deliverables.
Mangin, Eliamen A.	To be confirmed by the group	Specify agreed tasks and deliverables.
Somogba, Mares Susette V.	To be confirmed by the group	Specify agreed tasks and deliverables.

Responsibilities to allocate may include interface design and frontend development, backend and API development, database design, RAG/content-generation integration, code-execution and automated-assessment integration, machine-learning implementation and evaluation, testing, documentation, and coordination of research procedures. These are work areas to assign, not claims about completed individual contributions.

Budget Cost/Management Plan

The budget will estimate the costs required to develop, deploy, test, and document ByteQuest. The amounts below must be completed using current provider pricing, the expected duration of use, and the group's actual choices. Free-tier services should be listed as zero direct cost only if their limits are sufficient for the planned use.

Expense item	**Basis of estimate**	**Estimated cost (PHP)**
Web/application hosting	Selected provider and planned hosting period	To be determined
Database hosting	Selected database plan and duration	To be determined
LLM API usage	Expected number of requests/tokens and current pricing	To be determined
Code execution service or infrastructure	Chosen execution method and expected usage	To be determined

Internet connectivity	Project-related cost, if separately budgeted	To be determined
Printing and documentation	Estimated copies and pages	To be determined
Questionnaire validation or related expenses	Include only if a fee is required	To be determined
Contingency	If permitted by the adviser and based on the total estimate	To be determined
Total estimated budget	Calculate after the individual costs are confirmed	To be determined

The researchers will monitor actual expenses during development and compare them with the approved estimate. The final budget should distinguish one-time costs from recurring costs and should identify any services provided through free tiers, university resources, or existing personal equipment.

Ethical Considerations and Data Privacy

Before formal data collection, the researchers will seek the permissions required by the university and follow the applicable research procedures. Participation will be voluntary, and participants will be informed of the study's purpose, the information to be collected, and the intended use of their responses or programming-performance records. Required consent procedures will be followed.

Student responses and performance records will be handled as confidential research data. The researchers will limit access to the data to authorized members of the research team and will avoid including directly identifying information in analysis files unless it is necessary and authorized. Data will be stored and retained according to the approved study procedure and applicable institutional requirements, including relevant obligations under the Data Privacy Act of 2012 (Republic Act No. 10173). The researchers will confirm the university's specific requirements for consent, data storage, access, retention, and disposal before collecting data.

Submitted code will be executed only through the selected controlled environment. The system will apply appropriate restrictions to student submissions and will distinguish code-related outcomes from execution-service

or infrastructure failures when recording task results. AI-generated questions and challenges will be reviewed for correctness and relevance before use.